

Adriel Alfeus Hutabarat

Bekasi, Jawa Barat | +6281293586210 | adrielalfeus@gmail.com
<https://adriel.id> | <https://github.com/drithh>

EXPERIENCE

Software Engineer

Idea Comindo

June 2025 - Present

- Architected and built a **geospatial backend** using **PostGIS** for spatial operations and proximity analytics, with a **GraphQL API** powered by **NestJS**, leveraging **Redis**, **Dataloader**, and **context-based field filtering** for efficient and precise data access.
- Developed a **Python scraping platform** to automate the collection and ingestion of **Wikipedia administrative hierarchies**, **census and demographic datasets**, and **real-time satellite tracking** such as the ISS.
- Implemented an **AI-driven question answering** and **summarization** system using **Gemini** and **OpenAI** to generate automated insights from geospatial and demographic data.
- Built a **React** and **TailwindCSS** admin interface for user administration, usage logs, API key management, and dataset oversight.

Software Engineer

Itemgame

Nov 2024 - Aug 2025

- Built a **high-performance e-commerce** platform for game top-ups from the ground up, using **Astro** with **TypeScript** for a lightning-fast, static-first frontend and **Go** to power robust, scalable backend APIs.
- Integrated **NATS** for both efficient caching and real-time communication—leveraging its **KV store** to **reduce database load and latency**, and using its **job queues** to ensure seamless service coordination and responsive task processing.
- Optimized infrastructure for peak traffic by **containerizing all services** with **Docker** and deploying **CDN-optimized** builds via **Cloudflare**, enhancing performance, scalability, and deployment reliability.

Backend Engineer

PT Datasintesa Teknologi Nusantara

June 2024 - June 2025

- Develop and maintain a **BTS monitoring system for 5,800+ sites** across multiple consortiums, tracking **diverse KPI values** and handling high data payloads with **NestJS**, **TypeScript**, and **PostgreSQL**.
- Manage dozens of scheduled **Python** scripts (cron jobs) for data extraction from various BTS formats that uses **HTTP** or **SFTP** depending on the vendor.
- Handle asynchronous data exports between **NestJS** and **Python**, and integrate **BullMQ** for job queue management and **Elastic APM** for performance monitoring.

Software Engineer

Freelance

Aug 2024 - Sep 2024

- Pilates Reform
- Developed a Pilates website using **Next.js** and **MySQL**, incorporating features for **package purchases**, **class bookings**, **loyalty programs**, and **milestone tracking**.
 - Built an intuitive **admin panel for efficient data management**, including discount control, and integrated WhatsApp notifications to boost user engagement and communication.

- Wijaya Door
- Developed the Wisdoor portfolio website using **Next.js** and **PostgreSQL**, featuring an **interactive 3D door configurator** for user design and customization.
 - Integrated **Plasmic** to streamline UI design, **enabling efficient creation**, seamless updates, and **easy customization of the interface**.

Software Engineer

Sebelas Maret University School ERP Project

Sep 2022 - May 2024

- Software Engineer specializing in **Go** and **ReactJS**, developing a web platform for facilitating teacher-student interactions, managing tasks, grading assignments, student selection, and tracking academic progress. It also allows parents to view report cards and **make payments for school activities**.
- **Enhancing the web application** with **TypeScript**, **Tanstack Query**, and **Vite** for improved performance and developer experience. **Strengthened API documentation** and improved frontend-backend alignment through the **introduction of OpenAPI Specification (OAS)** with type safety, ensuring a more reliable and seamless development workflow.

Backend Engineer

Bootcamp Startup Campus

Aug 2022 - Dec 2022

- Top performer in the Backend Engineer Track, **contributing exceptional technical expertise** to the success of the final project.
- **Led the collaboration** between the backend and data science teams, ensuring seamless teamwork and driving the project to exceed expectations.

Frontend Engineer

P!NGFEST2021

Mar 2021 - Nov 2021

- **Designed and implemented a responsive Home UI** for both web and mobile for P!NGFEST2021, a major annual event for Himaster UNS, incorporating **animations and refined design elements** to create a seamless and visually appealing user experience.
- Worked closely with cross-functional teams to **optimize the website's performance, focusing on load times and technical efficiency**, while ensuring it was fully accessible and engaging for all users.

Staff Technology Department

Himpunan Mahasiswa Informatika UNS

Jan 2021 - Jan 2022

- **Crafted and delivered challenging coding problem sets** for IF Commit during IF Week, boosting student participation and technical growth.
- **Led the "Coding Task"** initiative for OSPRODI orientation, mentoring incoming Informatics students and building a strong foundation in programming skills.

EDUCATION

Sebelas Maret University

Bachelor of Computer Science (3.92/4.00)

- Active member of **Himpunan Mahasiswa Informatika** (Informatics Student Association)
- Lecture Assistant for **Programming Concepts** and **Web Programming Lab**
- Research Assistant specializing in **Federated Learning**

CORE COMPETENCIES

- Proficient in **Go**, **TypeScript**, **Python**, **JavaScript**, and **PHP**, with strong backend development experience
- Experienced in modular architecture, clean code practices, and **Agile methodologies** including Scrum and Kanban
- Skilled in **Google Cloud Platform (GCP)**, **Docker**, and setting up efficient CI/CD pipelines
- Hands-on with **NATS** and **Redis** for messaging, task queues, and key-value storage, and **PostgreSQL** for relational data
- Practical experience using **Elasticsearch for APM** and observability
- Strong **Git** knowledge with collaborative workflows, pull requests, and branching strategies